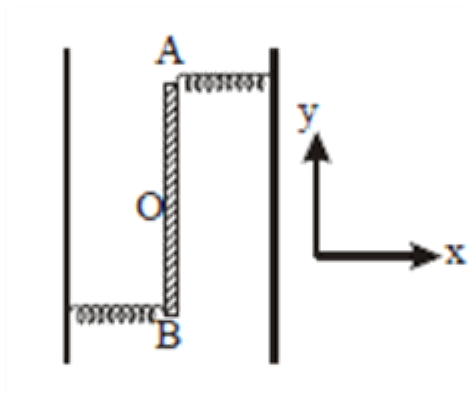
| Std 11 | **Lakshya**<br>**SUBJECT** : Physics<br>LOM, Friction, SHM, Thermodynamics | **Marks :** 100 |
|---|---|---|
| Weekly test | | **Time (mm:ss) :** 60:00 |

**1.** Two light identical springs of spring constant k are attached horizontally at the two ends of a uniform horizontal rod AB of length l and mass m. The rod is pivoted at its centre 'O' and can rotate freely in horizontal plane. The other ends of the two springs are fixed to rigid supports as shown in figure. The rod is gently pushed through a small angle and released. The frequency of resulting oscillation is




**a.** $\frac{1}{2\pi}\sqrt{\frac{6k}{m}}$

**b.** $\frac{1}{2\pi}\sqrt{\frac{2k}{m}}$

**c.** $\frac{1}{2\pi}\sqrt{\frac{k}{m}}$

**d.** $\frac{1}{2\pi}\sqrt{\frac{3k}{m}}$

**2.** A particle performs simple harmonic motion with amplitude A. If its speed is doubled at the instant when it is at distance $\frac{A}{3}$ from equilibrium position. The new amplitude of the motion is

**a.** $\sqrt{11}\ A$

**b.** $\frac{\sqrt{33}\ A}{2}$

**c.** $\frac{\sqrt{33}\ A}{3}$

**d.** None of these

**3.** A simple pendulum has a time period of $T_0$ on the surface of the earth. On another planet whose density is same that of earth but radius is 4 times then time period of this simple pendulum would be :

**a.** $T_0$

**b.** $\frac{T_0}{4}$

c. $\frac{T_0}{2}$

d. 4 $T_0$

**4.** A wooden cube (density of wood d) of side *l* floats in a liquid of density $\rho$ with its upper and lower surfaces horizontal. If the cube is pushed slightly down and released, it performs simple harmonic motion of period T. Then T is equal to

a. $2\pi\sqrt{\frac{\ell\rho}{(\rho-d)\,g}}$

b. $2\pi\sqrt{\frac{\ell d}{\rho g}}$

c. $2\pi\sqrt{\frac{\ell\rho}{dg}}$

d. $2\pi\sqrt{\frac{\ell d}{(\rho-d)\,g}}$

**5.** A particle at the end of a spring executes simple harmonic motion with a period $t_1$, while the corresponding period for another spring is $t_2$. If the period of oscillation with the two springs in series is T, then

a. $T = t_1 + t_2$

b. $T^2=t_1^2+t_2^2$

c. $T^{-1}=t_1^{-1}+t_2^{-1}$

d. $T^{-2}=t_1^{-2}+t_2^{-2}$

**6.** For a body executing SHM
A. potential energy is always equal to its kinetic energy.
B. average potential and kinetic energy over any given time interval are always equal. C. sum of the kinetic and potential energy at any point of time is constant.
D. average kinetic energy in one time period is equal to average potential energy in one time period.
Choose the most appropriate option from the options given below.

a. (C) and (D)

b. Only (C)

c. (B) and (C)

d. Only (B)

**7.** The bob of a simple pendulum is a spherical hollow ball filled with water. A plugged hole near the bottom of the oscillating bob gets suddenly unplugged. During observation, till water is coming out, the time period of oscillation would

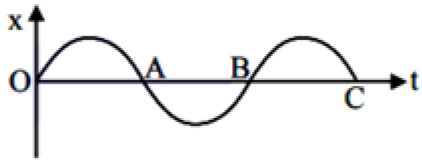
a. first decrease and then increase to the original value

b. first increase and then decrease to the original value

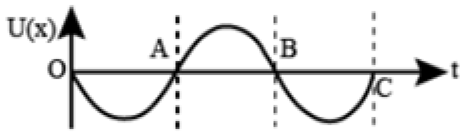
c. increase towards a saturation value

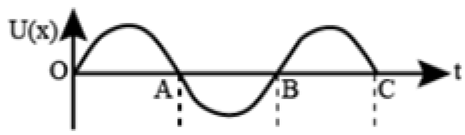
**d.** remain unchanged

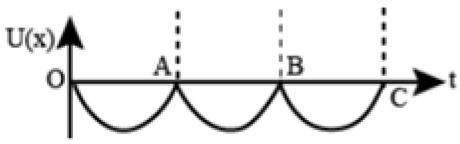
**8.** The variation of displacement with time of a particle executing free simple harmonic motion is shown in the figure

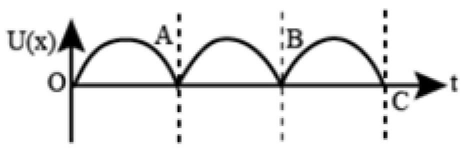



The potential energy U(x) versus time (t) plot of the particle is correctly shown in figure :

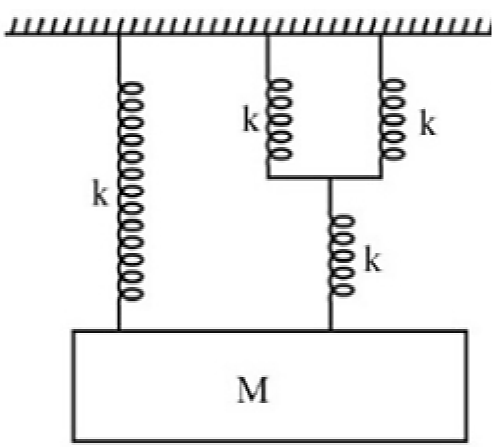
**a.**



**b.**



**c.**



**d.**



**9.** The time period of simple harmonic motion of mass M in the given figure is $\pi\sqrt{\frac{\alpha M}{5K}}$, where the value of α is _____



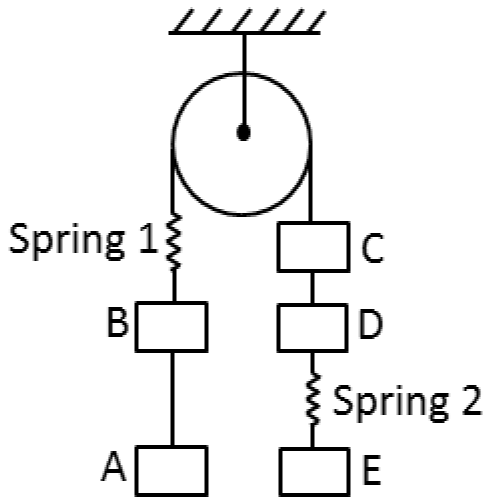

**a.** 12

**b.** 24

**c.** 6

**d.** 10

**10.** The system shown in the figure is initially in equilibrium.




| Column-I | Column-II |
|---|---|
| (A) Just after the Spring 2 is cut, the block D | (P) Accelerates up |
| (B) Just after the Spring 2 is cut, the block C | (Q) Accelerates down |
| (C) Just after the Spring 2 is cut, the block A | (R) Momentarily at rest |
| (D) Just after the Spring connecting A and B is cut, the block D | (S) Moves up with accelerationg |

**a.** A-P ; B-Q ; C-R ; D-S

**b.** A-S ; B-R ; C-Q ; D-P

**c.** A-P ; B-P ; C-R ; D-R

**d.** A-Q ; B-P ; C-R ; D-S

**11.** A given object takes n times more time to slide down a $45^{o}$ rough inclined plane as it takes to slide down a perfectly smooth $45^{o}$ incline. The coefficient of kinetic friction between the object and the incline is

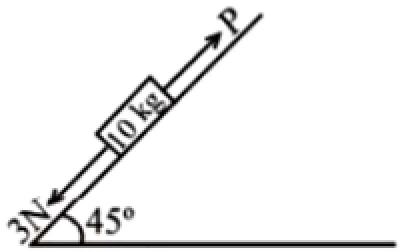
**a.** $\sqrt{1-\frac{1}{n^2}}$

**b.** $1-\frac{1}{n^2}$

**c.** $\frac{1}{2-n^2}$

**d.** $\sqrt{\frac{1}{1-n^2}}$

**12.** A block of mass 10 kg is kept on a rough inclined plane as shown in the figure. A force of 3 N is applied on the block. The coefficient of static friction between the plane and the block is 0.6. What should be the minimum value of force P, such that the block does not move downward ? (Take g = 10 Acc. Over Quota ) The minimum value of P such that block does not move downward is Acc. Over Quota N. find x.



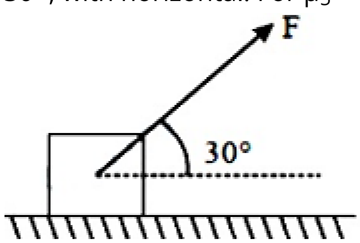

**a.** 20

**b.** 10

**c.** 15

**d.** 5

**13.** As shown in the figure a block of mass 10kg lying on a horizontal surface is pulled by a force F acting at an angle 30° , with horizontal. For $\mu_5 = 0.25$, the block will just start to move for the value of F : [Given g = $10 ms^{-2}$ ]




**a.** 33.3N

**b.** 25.2N

**c.** 20N

**d.** 35.7N

**14.** A particle of mass m is moving in a straight line with momentum p. Starting at time t = 0, a force F = kt acts in the same direction on the moving particle during time interval T so that its momentum changes from p to 3p. Here k is a constant. The value of T is

**a.** $2\sqrt{\frac{p}{k}}$

**b.** $\sqrt{\frac{2k}{p}}$

**c.** $\sqrt{\frac{2p}{k}}$

**d.** $2\sqrt{\frac{k}{p}}$

**15.** A spaceship in space sweeps stationary interplanetary dust. As a result, its mass increases at a rate $\frac{dM\,(t)}{dt} = bv^2\,(t)$ , where v(t) is its instantaneous velocity. The instantaneous acceleration of the satellite is

**a.** $-\,bv^3\,(t)$

**b.** $\frac{bv^3(t)}{M\,(t)}$

**c.** $-\frac{2bv^3(t)}{M\,(t)}$

**d.** $-\frac{bv^3(t)}{2M\,(t)}$

**16.** A mass of 10kg is suspended by a rope of length 4m, from the ceiling. A force F is applied horizontally at the mid-point of the rope such that the top half of the rope makes an angle of 45° with the vertical. Then F equals (Take g = 10 m/$s^2$ and rope to be massless)

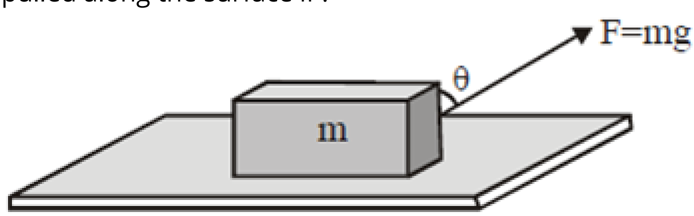
**a.** 100N

**b.** 90N

**c.** 75N

**d.** 70N

**17.** A force F is acting on a block of mass m. Coefficient of friction between block & surface is $\mu$. The block can be pulled along the surface if :-



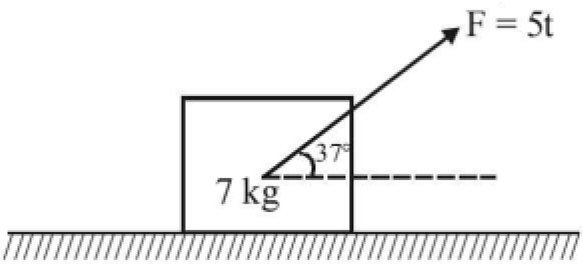

**a.** $\tan\ \theta \geq \mu$

**b.** $\cos\ \theta \geq \mu$

**c.** $\tan\frac{\theta}{2} \geq \mu$

**d.** $\cot\frac{\theta}{2} \geq \mu$

**18.** A block of 7 kg is placed on a rough horizontal surface and is pulled through a variable force F (in N) = 5 t, where t is time in second, at an angle of $37^0$ with the horizontal as shown in figure, The coefficient of static friction of the block with the surface is one. If the force starts acting at t = 0 s, the time at which the block starts to slide is $t_0$ sec. Find the value of $t_0/2$ in sec. ( g = 10 m/$s^2$ and cos $37^0$ = $\frac{4}{5}$)




**a.** 3

**b.** 4

**c.** 5

**d.** 6

**19.** If for hydrogen $C_p - C_v = m$ and for nitrogen $C_p - C_v = n$, where $C_p$ and $C_v$ refer to specific heats per unit mass respectively at constant pressure and constant volume, the relation between m and n is (molecular weight of hydrogen = 2 and molecular weight of nitrogen = 14)

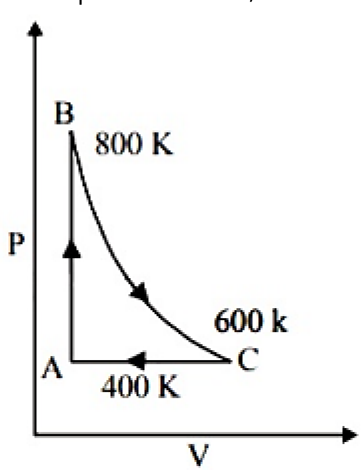
**a.** n = 14m

**b.** n = 7m

**c.** m = 7n

**d.** m = 14n

**20.** One mole of a diatomic ideal gas undergoes a cyclic process ABC as shown in figure. The process BC is adiabatic. The temperatures at A, B and C are 400 K, 800 K and 600 K respectively. Choose the correct statement




**a.** The change in internal energy in whole cyclic process is 250 R.

b. The change in internal energy in the process CA is 700 R

c. The change in internal energy in the process AB is - 350 R.

d. The change in internal energy in the process BC is – 500 R

**21.** The pressure and volume of an ideal gas are related as $PV^{3/2}$ = K (Constant). The work done when the gas is taken from state A($P_1$, $V_1$, $T_1$) to state B($P_2$, $V_2$, $T_2$) is :

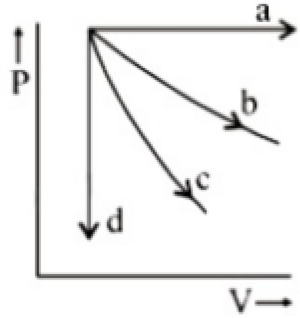
a. $2(P_1V_1 - P_2V_2)$

b. $2(P_2V_2 - P_1V_1)$

c. $2\left(\sqrt{P_1}V_1 - \sqrt{P_2}V_2\right)$

d. $2\left(P_2\sqrt{V_2} - P_1\sqrt{V_1}\right)$

**22.** The given diagram shows four processes i.e., isochoric, isobaric, isothermal and adiabatic. The correct assignment of the processes, in the same order is given by :



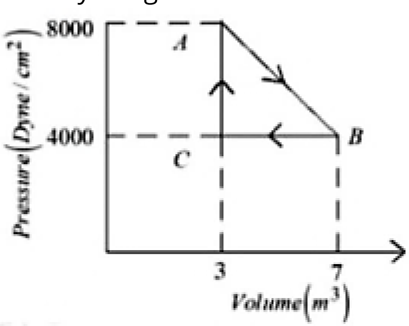

a. a d b c

b. d a c b

c. a d c b

d. d a b c

**23.** A thermodynamic system is taken from an original state A to an intermediate state B by a linear process as shown in the figure. It's volume is then reduced to the original value from B to C by an isobaric process. The total work done by the gas from A to B and B to C would be :




a. 33800 J

b. 2200 J

c. 800 J

d. 1200 J

**24.** The ratio of work done by an ideal monoatomic gas to the heat supplied to it in an isobaric process is:

a. $\frac{3}{5}$

b. $\frac{2}{3}$

c. $\frac{3}{2}$

d. $\frac{2}{5}$

**25.** One mole of an ideal monoatomic gas is compressed isothermally in a rigid vessel to double its pressure at room temperature, $27^{0}$C. The work done on the gas will be :

a. 300 R

b. 300 R ln 6

c. 300 R ln 2

d. 300 R ln 7